

Full-window source products and the retained class: finite quotient comparisons in the original arithmetic metric

Split-Zero research programme

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The user-supplied continuation [1] proposes an evaluation of the two complete source-response products and the four-cutoff terminal-class return. We prove these evaluations below, retaining the covariance correction as an exact finite sum. In particular, their proof does not require a new uniform exponential estimate for that covariance. The already established finite original-source multiplication and exterior estimates suffice, at every fixed primary multiplicity. We also give the exact receiving identity for the same class in the original observation quotient. Statements about that observation keep its actual projection angle and complex responses.

1 Original packet, attained class and source energies

Fix the actual quartet data of the preceding calculation:

$$\begin{aligned} \rho &= \tfrac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad m_0 \geq 1, \\ k &\equiv 1 \pmod{4}, \quad k \geq 5, \quad e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2. \end{aligned} \tag{FW1}$$

Keep the full quartet factor h in 2ξ , its full arithmetic unit and the literal measure and physical coordinate

$$w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi), \quad dm = w_h^{*k}(y) dy, \quad S = c + iy, \quad c = k/2.$$

The original complete polynomial and terminal class are

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, \quad E = \mathbb{C}[S]/(\chi), \\ \lambda &= k\rho = c + d + it, \quad d = k\delta, \quad t = k\gamma, \quad z = t - id, \quad R = |z| = k\sqrt{\delta^2 + \gamma^2}, \\ \chi_\lambda &= \chi/(S - \lambda)^e, \quad b_\lambda = 1/\chi_\lambda(\lambda), \quad v = [b_\lambda \chi/(S - \lambda)]. \end{aligned} \tag{FW2}$$

No root difference, repeated terminal power or arithmetic scalar has been changed. Write $Q(y) = i^{-q}\chi(c + iy)$. Conjugation and negation preserve its full root multiset, and no root is real. Thus Q is real, even, strictly positive on the real line, and $Q(z) = Q(\bar{z}) = 0$. Put $dW = Q^2 dm$. Its real monic orthogonal polynomials and squared norms are U_j, ν_j . Define

$$C_j(z) = \int \frac{U_j(y)}{z - y} dW(y), \quad H_r(z) = \min_{P \in \mathbb{C}[y]_{<r}} \int \left| \frac{1}{z - y} - P(y) \right|^2 dW(y). \tag{FW3}$$

The minimum is attained by $\sum_{j < r} C_j(z) U_j/\nu_j$. Multiplication by $-b_\lambda i^{q-1} Q$ maps this affine fibre exactly to all representatives of v through degree $N = q - 1 + r$; its inverse divides by that same specified nonzero factor. This gives

$$E_N = v^* G_N v = |b_\lambda|^2 H_r, \quad G_N = (J_N H_N^{\text{source}, -1} J_N^*)^{-1}, \quad H_r - H_{r+1} = |C_r|^2/\nu_r. \tag{FW4}$$

Here J_N is the original remainder map and H_N^{source} is the Gram of $1, S, \dots, S^N$ in dm . Every original relation is included in its minimum. The finite differences in (FW4) need no infinite orthogonal expansion.

Let $p_n(S)$ be the original S -monic source polynomial, $\omega_n = \|p_n\|_m^2$, and $b_n = [p_n]$. In the three columns (b_N, b_{N+1}, v) put $F_{ij} = W_i^* G_N W_j$ and

$$p_{1,N} = \frac{|F_{13}|^2}{F_{11}E_N}, \quad p_{2,N} = \frac{|F_{23}|^2}{F_{22}E_N}, \quad \Delta_r = \omega_{q+r}/\nu_r.$$

The full monic relation of degree $q+r$ is $\chi\pi_r$, where $\pi_r(c+iy) = i^r U_r(y)$. Its difference from p_{q+r} has degree at most $q+r-1$, represents b_{q+r} , and is orthogonal to every preceding relation. Squaring its norm gives $F_{22} = \nu_r - \omega_{q+r}$. The same monic projection in degree $q+r-1$ gives

$$p_{2,q-1+r} = \frac{H_r - H_{r+1}}{(1 - \Delta_r)H_r}, \quad p_{1,q-1+r} = \frac{\Delta_{r-1}}{1 - \Delta_{r-1}} \left(\frac{H_{r-1}}{H_r} - 1 \right) \quad (r \geq 1). \quad (\text{FW5})$$

At $r=0$ the unchanged first formula is $p_{1,q-1} = \omega_{q-1}/H_0$. These are precisely the full minimum projections in RBE12–19 [7]; the changes of variable above explain every factor of i .

Lemma 1 (Finite residual Wronskian). *For $r \geq 1$,*

$$H_r = \frac{\text{Im}(C_r \overline{C_{r-1}})}{d\nu_{r-1}} > 0. \quad (\text{FW6})$$

Proof. Evenness and monicity give $U_{j+1} = yU_j - (\nu_j/\nu_{j-1})U_{j-1}$. Integrating the recurrence gives $C_1 = zC_0 - \nu_0$ and $C_{j+1} = zC_j - (\nu_j/\nu_{j-1})C_{j-1}$ for $j \geq 1$. Since $\text{Im } C_0 = dH_0$, the right side for $r=1$ is $H_0 - |C_0|^2/\nu_0 = H_1$. At each later step its difference is $-|C_j|^2/\nu_j$, exactly (FW4). Positivity follows because the rational function cannot equal a polynomial on an interval. $\square$

The recurrence, simple real zeros and interlacing used below are the classical orthogonal-polynomial results recorded in DLMF [2, §18.2(iv), (vi), (vii)]. The residual identities themselves have just been proved for the original arithmetic measure.

2 Finite control of the central interval and rational minima

The original CAI16–25 estimates [6] state, for every finite degree n , with $M = 21 + 4m_0$,

$$\frac{a_k}{[1 + 4(n+M)^2]^M} \|P\|_\sigma^2 \leq \|P\|_m^2 \leq A_k \|P\|_\sigma^2, \quad \|yP\|_m \leq C_{\text{mult}}(h)(2n + \lfloor k/3 \rfloor) \|P\|_m. \quad (\text{FW7})$$

Here $\sigma = |\Gamma(1/4 + iy/2)|^2/(2\pi)$, of mass $\sqrt{2\pi}$; the positive constants $a_k, A_k, C_{\text{mult}}$ retain the source masses and quartet from CAI16 and CAI22. In particular $\log(A_k/a_k) = O_h(k + \log k)$. These estimates hold for complex polynomials, not merely real monomials. We use $n = 2q+2$ and denote the two comparison constants by ℓ, u , and $\Lambda = \log(u/\ell)$. For Gamma, $\|yP\|_\sigma \leq 2(n+1) \|P\|_\sigma$, from its exact three-term recurrence. No arithmetic metric is replaced by this reference.

Choose a fixed $\epsilon_* \leq 2^{-16}$ with ϵ_*^2 below the support of $\rho_{2,\pi}$, put $Y = \epsilon_* q$, $D = q+2$, and impose $R \leq Y/4$. With the original Gamma lower-envelope constant $c_G > 0$, set

$$\eta = \frac{u}{\ell} \frac{\sqrt{2\pi}}{c_G} \frac{(D+1)^2 \sqrt{1+2q}}{q}, 100^D e^{\pi q} \left(\frac{Y^2 + R^2}{q^2 - R^2} \right)^q. \quad (\text{FW8})$$

Then, for every $\deg P \leq D$,

$$\int_{|y| \leq Y} |P|^2 dW \leq \eta \|P\|_W^2, \quad \eta \leq e^{-q \log 16 + O_h(k + \log q)}. \quad (\text{FW9})$$

To verify this, the shifted Legendre basis on $[q, 2q]$ gives

$$\sup_{|y| \leq Y} |P(y)|^2 \leq (D+1)^2 100^D q^{-1} \int_q^{2q} |P|^2;$$

its argument belongs to $[-4, 4]$, and the defining Legendre formula bounds each degree- j basis value by 10^j times its normalization. Opposite-root pairs give $|Q|^2 \leq (Y^2 + R^2)^q$ centrally and $|Q|^2 \geq (q^2 - R^2)^q$ on $[q, 2q]$. The central measure has mass at most $u\sqrt{2\pi}$, by (FW7) for 1; the full denominator is at least $\ell \int_q^{2q} |PQ|^2 \sigma$. The Gamma lower density there is $c_G(1+2q)^{-1/2} e^{-\pi q}$. These statements prove (FW8) without assuming a pointwise comparison of arithmetic and Gamma densities. Finally $100e^\pi(17\epsilon_*^2/16)/(1 - \epsilon_*^2/16) < 1/16$, which proves the last assertion. Both complementary regions remain in the original forms.

Put $W^\circ = 1_{|y| > Y} W$, and use a superscript $\circ$ for its polynomials, norms and residuals. Take $C = C_{\text{mult}}(h)[2(2q+1) + \lfloor k/3 \rfloor]$ and $\xi = \eta(C+R)^2/d^2$. For $0 \leq r \leq q+2$,

$$(1 - \xi)H_r \leq H_r^\circ \leq H_r, \quad |T_r - T_r^\circ| \leq \frac{\xi}{1 - \xi}, \quad T_r = 1 - H_{r+1}/H_r. \quad (\text{FW10})$$

The second statement uses $r \leq q+1$. Indeed each candidate has numerator $Z = 1 - (z - y)P$ of degree at most r . On the central interval its squared norm is at most $d^{-2} \int |Z|^2 dW$ restricted there. Apply (FW9) to Z . The polynomial $F = QZ/(z - y)$ exists because $Q(z) = 0$, and has degree at most $q + r - 1$. Moreover $\|Z\|_W = \|(z - y)F\|_m \leq (C + R)\|F\|_m$. This proves the candidate-wise lower bound and hence the attained one. Writing $H_r^\circ = a_r H_r$, $1 - \xi \leq a_r \leq 1$, proves the ratio bound in (FW10). Degree $2q + 2$ in (FW7) includes its final output.

For $j \leq q + 1$ define

$$M_j^\circ = \int \frac{(U_j^\circ)^2}{y^2} dW^\circ, \quad \hat{C} = \frac{C_{\text{mult}}(h)[2(2q+1) + \lfloor k/3 \rfloor]}{\sqrt{1 - \eta}}.$$

Multiplication on QU_j and (FW9) bound $\|yU_j^\circ\|_{W^\circ} \leq \hat{C}\|U_j^\circ\|_{W^\circ}$. Cauchy-Schwarz applied to U_j/y and yU_j , and the pointwise $|y| > Y$, now give

$$\hat{C}^{-2} \leq M_j^\circ/\nu_j^\circ \leq Y^{-2}, \quad \nu_j^\circ/\nu_{j-1}^\circ \leq \hat{C}^2. \quad (\text{FW11})$$

The second inequality is the minimum monic norm applied to yU_{j-1} .

3 Consecutive amplitudes and exact product cancellation

Write $U_{2j}^\circ(y) = P_j(y^2)$, $U_{2j+1}^\circ(y) = yT_j(y^2)$. Orthogonality yields

$$\int T_j(y^2) dW^\circ = \frac{\nu_{2j}^\circ}{P_j(0)}, \quad M_{2j+1}^\circ = \frac{T_j(0)\nu_{2j}^\circ}{P_j(0)}, \quad P_{j+1}(0) = -\frac{\nu_{2j+1}^\circ}{\nu_{2j}^\circ} P_j(0). \quad (\text{FW12})$$

For the first identity, $P_j - P_j(0)$ is x times a degree- $j - 1$ polynomial, so its product with T_j integrates to zero; while $\int P_j T_j = \int P_j^2 = \nu_{2j}^\circ$. For the second, $T_j - T_j(0)$ is likewise x times a lower polynomial, orthogonal to T_j in the x -weighted measure. The last equality is the monic recurrence at zero. Quadratic transformation places every positive squared zero above Y^2 , and the two consecutive zero lists interlace.

Put $\varrho = R^2/Y^2$. Orthogonality of $(U_j(z) - U_j(y))/(z - y)$ against U_j proves the exact identity

$$U_j^\circ(z)C_j^\circ(z) = \int \frac{(U_j^\circ(y))^2}{z - y} dW^\circ = -zM_j^\circ A_j(z), \quad |A_j(z) - 1| \leq \frac{\varrho}{1 - \varrho}. \quad (\text{FW13})$$

The last step pairs $y, -y$: A_j is the probability-weighted average of $(1 - z^2/y^2)^{-1}$ with weight $(U_j^\circ)^2/y^2$. For interlacing positive squared zeros, the signed counting function has absolute value

at most one. Integrating $\partial_x \log(1 - z^2/x)$ from Y^2 to infinity therefore bounds the logarithm of the consecutive root-product ratio by $\int_{Y^2}^\infty R^2/[x(x - R^2)] dx = -\log(1 - \varrho)$. This proves the complex ratio estimate, including its phase and without a degree factor. In conjunction with (FW12)–(FW13) it gives

$$\frac{C_r^\circ}{C_{r-1}^\circ} = \begin{cases} -z(M_r^\circ/\nu_{r-1}^\circ)(1 + v_r), & r \text{ even}, \\ z^{-1}(\nu_{r-1}^\circ/M_{r-1}^\circ)(1 + v_r), & r \text{ odd}, \end{cases} \quad |v_r| \leq 8\varrho \quad (\varrho \leq 1/16). \quad (\text{FW14})$$

For explicit verification of the constant, each A -ratio differs from one by at most $2\varrho/(1 - 2\varrho)$, and the product ratio by at most $\varrho/(1 - \varrho)$; multiplying the factors gives a bound smaller than 8ϱ on the stated interval.

Define positive finite quantities

$$b_r = \begin{cases} R^2 M_r^\circ / \nu_r^\circ, & r \text{ even}, \\ (\nu_{r-1}^\circ)^2 / (\nu_r^\circ M_{r-1}^\circ), & r \text{ odd}, \end{cases} \quad a_* = 32(1 + R/d)\varrho + \frac{\xi}{1 - \xi} \frac{\hat{C}^2}{R^2}. \quad (\text{FW15})$$

Impose the explicit eventual guards $\eta, \xi < 1/4$, $\varrho \leq 1/16$, $32(1 + R/d)\varrho \leq 1/4$, and $a_* \leq 1/4$. Then

$$T_r = b_r(1 + \varepsilon_r), \quad |\varepsilon_r| \leq a_*, \quad p_{2,q-1+r} = \frac{b_r(1 + \varepsilon_r)}{1 - \Delta_r}, \quad 0 \leq r \leq q + 1. \quad (\text{FW16})$$

For $r \geq 1$ substitute (FW14) in (FW6). The expression is

$$T_r^\circ = \frac{d\nu_{r-1}^\circ |C_r^\circ/C_{r-1}^\circ|^2}{\nu_r^\circ \text{Im}(C_r^\circ/C_{r-1}^\circ)}.$$

The relative error of the imaginary part is at most $8(R/d)\varrho$; the numerator error is at most $16\varrho + 64\varrho^2$. Their ratio is within $32(1 + R/d)\varrho$ of one on the guard. For $r = 0$, use $C_0 = -zM_0A_0$, $H_0 = \text{Im } C_0/d$, which has the same bound. Finally (FW11) gives $b_r \geq R^2/\hat{C}^2$ for even r and $b_r \geq Y^2/\hat{C}^2 \geq R^2/\hat{C}^2$ for odd r . Thus (FW10) gives exactly the remaining relative allowance in a_* .

Let $w_0 = w_q = 1$, $w_r = 2$ for $0 < r < q$, so $\sum w_r = 2q$. Define the exact positive covariance contribution

$$D_k = \sum_{r=0}^q w_r [-\log(1 - \Delta_r)], \quad \mathcal{A}_i = -\sum_{r=0}^q w_r \log p_{i,q-1+r}. \quad (\text{FW17})$$

Equation (FW12) and the two definitions of b imply $b_{2j}b_{2j+1} = R^2\nu_{2j}^\circ/\nu_{2j+1}^\circ$ and $|U_q^\circ(0)| = \prod_{j < q/2} \nu_{2j+1}^\circ/\nu_{2j}^\circ$. Taking the literal endpoint-weighted sum therefore proves

$$\mathcal{A}_2 = 2 \log |U_q^\circ(0)| - 2q \log R - \log \frac{M_q^\circ/\nu_q^\circ}{M_0^\circ/\nu_0^\circ} - D_k + \mathcal{E}_k, \quad |\mathcal{E}_k| \leq 4qa_*. \quad (\text{FW18})$$

Every inverse moment except the displayed endpoint ratio has cancelled. Its logarithm is bounded by $2 \log(\hat{C}/Y)$.

A finite covariance bound suffices. Let $L_{\text{PT}} = \delta q(k + 1)/2$ and $B = C_{\text{mult}}(h)^2(4q + \lfloor k/3 \rfloor)^2$. The original exterior determinant bound gives $\kappa_N \geq L_{\text{PT}}$ at every $N \geq q - 1$. Direct substitution of the two monic boundary norms gives $\kappa_{q-1}^2 = (\omega_q/\omega_{q-1})(\Delta_0^{-1} - 1)$ and $\kappa_N^2 = (\omega_{N+1}/\omega_N)(1 - \Delta_{r-1})(\Delta_r^{-1} - 1)$ at $r \geq 1$. The source multiplication bound controls the recurrence coefficient by B . Consequently

$$0 < \Delta_r \leq \frac{B}{B + L_{\text{PT}}^2}, \quad 0 \leq D_k \leq 2q \log(1 + B/L_{\text{PT}}^2) = O_h(q/k^2) = O_h(k). \quad (\text{FW19})$$

For $r = q + 1$, enlarge $4q$ in B to $4q + 2$; the same proof applies. This derives exactly the needed estimate from PT7, PT23 and CAI25, as in RBE26. There is no use of a conjectured covariance limit. Moreover $qa_* = O_h(k^2/q) + e^{-cq + O_h(k + \log q)}$ and $\hat{C}/Y = O_h(1)$. Thus the exact correction in (FW18) fits in $O_h(k + \log q)$ at every fixed m_0 .

4 Two quotient metrics evaluate the polynomial value

Push W° forward under the literal $x = (y/q)^2$, preserving its full mass. Put $n = q/2$, and write $U_q^\circ(y) = q^q \hat{P}_n((y/q)^2)$. For both this arithmetic measure and the Gamma reference with the same full Q and gap, put

$$D_n^\mu(a) = \det\left[\int x^{i+j+a} d\mu\right]_{i,j < n}, \quad f_\mu(a) = \log D_n^\mu(a).$$

Solving the monic orthogonality equations for the constant coefficient by Cramer's rule yields $|\hat{P}_n(0)| = D_n^m(1)/D_n^m(0)$. All determinants are positive. Their squared-Vandermonde integral, with its factor $1/n!$, makes $f_\mu''(a)$ the variance of $\sum \log x_j$. Differentiation is justified by the positive lower endpoint and exponential tail. Hence

$$\frac{f_m(0) - f_m(-2)}{2} \leq \log |\hat{P}_n(0)| \leq \frac{f_m(2) - f_m(0)}{2}. \quad (\text{FW20})$$

Here is the full metric comparison. On $V = \mathbb{C}[x]_{<n}$ and xV , their common subspace is $x\mathbb{C}[x]_{<n-1}$. In each determinant factor it out by a Schur complement. The common Gram is identical within each measure and cancels. The remaining factors are the attained norms of 1 and x^n modulo this same subspace. An inequality $\ell^\circ \|F\|_\Gamma^2 \leq \|F\|_m^2 \leq u^\circ \|F\|_\Gamma^2$ passes to each affine minimum. The ratio of the two quotient factors is therefore between ℓ°/u° and u°/ℓ° times its Gamma value. No determinant dimension appears. Applied to (FW7)–(FW9), this gives

$$|[f_m(2) - f_m(0)] - [f_\Gamma(2) - f_\Gamma(0)]| \leq \Lambda^\circ, \quad \ell^\circ = (1 - \eta)\ell, \quad u^\circ = u/(1 - \eta_\Gamma), \quad \Lambda^\circ = \log(u^\circ/\ell^\circ). \quad (\text{FW21})$$

The Gamma central fraction η_Γ is (FW8) with $u/\ell = 1$. Every polynomial here, after multiplication by Q and the specified substitution, has degree at most $2q$.

For the negative slope use $V, x^{-1}V$, with common subspace $\mathbb{C}[x]_{<n-1}$. To compare their entire sum, write each rational element as F/x with $\deg F \leq n$. In either gapped measure, the full polynomial multiplication bound, including Q , yields $\|yF\|_\mu \leq B_\mu \|F\|_\mu$, with $B_m, B_\Gamma = O_h(q)$ explicitly given by their degree- $2q$ constants divided by $\sqrt{1 - \eta_\mu}$. Jensen's inequality for $t \mapsto t^{-2}$ in the probability measure $|F|^2 d\mu / \|F\|_\mu^2$ gives

$$(q/B_\mu)^4 \|F\|_\mu^2 \leq \|F/x\|_\mu^2 \leq \epsilon_*^{-4} \|F\|_\mu^2.$$

Combining these two estimates with (FW21)'s polynomial comparison gives a rational logarithmic width

$$\Lambda^{\text{rat}} = \Lambda^\circ + 4 \log \frac{B_m B_\Gamma}{\epsilon_*^2 q^2} = O_h(k + \log q).$$

The identical common-Gram cancellation proves

$$|[f_m(0) - f_m(-2)] - [f_\Gamma(0) - f_\Gamma(-2)]| \leq \Lambda^{\text{rat}}. \quad (\text{FW22})$$

In particular the following finite interval has completely specified Gamma endpoints:

$$\frac{f_\Gamma(0) - f_\Gamma(-2) - \Lambda^{\text{rat}}}{2} \leq \log |\hat{P}_n(0)| \leq \frac{f_\Gamma(2) - f_\Gamma(0) + \Lambda^\circ}{2}. \quad (\text{FW23})$$

5 Gamma slopes, both source products and the arithmetic correction

Write $L_{\text{eq}} = \int \log x d\rho_{2,\pi}$ and $F = -\min \mathcal{E}_{2,\pi}$, with the original equilibrium conventions. The complete Gamma calculation AG10–18 [8] retains both complementary integration regions, all

roots and the fixed potential $V(x) = \pi\sqrt{x} - 2\log x$. At $n = q/2$ the exact density factors for $a = -2, 0, 2$ are

$$e^{-nV(x)+W_{a,k}(x)}, \quad W_{a,k} = (a - 3/4)\log x - \tau_k/x, \quad \tau_k = \frac{(\gamma^2 - \delta^2)k(k+2)}{3q}. \quad (\text{FW24})$$

On the fixed positive interval the complete root remainder has supremum $O_h(k^4/q^3)$, so its determinant cost is $O_h(k^4/q^2) = O_h(1)$. The Gamma-density remainder costs $O(q^{-1})$. Outside that interval the complete-form bounds of AG11–14 are exponentially small. For the three fixed powers here the same bounds apply: on the retained gap $x^{-2} \leq \epsilon_*^{-4}$, while at infinity x^2 adds only two fixed degrees to the polynomial estimate. The two additional degrees were included in (FW7). Constants and small-cutoff guards can thus be enlarged before, not after, the determinant comparison.

Borot and Guionnet’s one-cut analytic-statistic theorem [3, Propositions 1.2 and 5.2], in the monic determinant convention proved in AG18, gives

$$\log D_n[e^{-nV+W}] - \log D_n[e^{-nV}] = n \int W d\rho_{2,\pi} + O_W(1).$$

Its hypotheses hold on that fixed positive interval: AG6 gives a positive one-cut density, nonzero edge factors and a strict exterior effective-potential gap. Analyticity at zero is not invoked. Uniformity in the actual bounded τ_k follows by convexity: subtract the displayed linear statistic from the logarithm and bound this convex function at $-2T, 0, 2T$, with $T > 1 + \sup |\tau_k|$. Convexity bounds it above on $[-T, T]$ and below by expressing zero as a convex combination of an interior point and the opposite endpoint. This gives a common finite bound, separately for each of the three fixed powers. We have therefore proved

$$f_\Gamma(2) - f_\Gamma(0) = qL_{\text{eq}} + O_h(\log q), \quad f_\Gamma(0) - f_\Gamma(-2) = qL_{\text{eq}} + O_h(\log q). \quad (\text{FW25})$$

Equations (FW23)–(FW25), including the exact scaling of the monic polynomial, give

$$\log |U_q^\circ(0)| = q \log q + (q/2)L_{\text{eq}} + O_h(k + \log q). \quad (\text{FW26})$$

Theorem 2 (Complete original source-response products). *At every fixed original multiplicity, as $k \rightarrow \infty$ along (FW1),*

$$\mathcal{A}_2 = 2q \log \frac{q}{k\sqrt{\delta^2 + \gamma^2}} + qL_{\text{eq}} + O_h(k + \log q), \quad (\text{FW27})$$

$$\mathcal{A}_1 = J_k^{\text{action}} - 2q \log(qk\sqrt{\delta^2 + \gamma^2}) - qL_{\text{eq}} + O_h(k + \log q). \quad (\text{FW28})$$

Proof. (FW18), (FW19) and (FW26) prove the first equality. For the second, the original full-action boundary identity RBE3–7 and EC give $p_{1,N}p_{2,N} = |\mathbf{c}_N|^2/\kappa_N^2$ and $\mathbf{c}_N = d + i(-1)^r(t - \varepsilon_N)$ with $0 < \varepsilon_N = O_h(k^3/q^2)$ throughout the window. Thus $\log |\mathbf{c}_N| = \log R + O_h(k^2/q^2)$, retaining its signed phase before taking its modulus. The original action is $J_k^{\text{action}} = 2 \sum w_r \log \kappa_{q-1+r}$. Hence exactly

$$\mathcal{A}_1 + \mathcal{A}_2 = J_k^{\text{action}} - 4q \log R + O_h(k^2/q).$$

Subtract (FW27). All errors are $o_h(q/\log q)$ because $q = [1 + k(m_0 - 1)](k+1)^2$ with fixed m_0 . $\square$

In the unchanged constants of AG26–29, put $C_\Gamma = 2L_{\text{eq}} - 8\log 2 + 4$, $C_B = 9 - 8\log 2 + F$, $C_\sigma = F + 6 - 3L_{\text{eq}}/2 - 6\log 2 + 2\log \pi$, $A_J = C_\sigma + 8\log 2 - 4 - (4m_0 - 2)C_\Gamma$ and $r_h = 2(\gamma^2 - \delta^2)M_{-1}/3$, where $M_{-1} = \int x^{-1} d\rho_{2,\pi}$. Retain $I_h = \int_0^{\pi/2} [(\log M_h)'(\theta)]^2 d\theta$ with $M_h(\theta) = \int e^{\theta y} w_h(y) dy$, and $c_\zeta = 2\gamma_E - \log(2\pi)$. Substitution of the proved action AG29, with its SR1–48 finite remainder, in (FW28) yields

$$\begin{aligned} \mathcal{A}_1 = & C_B q^2 + 2q \log(q/k) + (A_J - L_{\text{eq}} - 2 \log \sqrt{\delta^2 + \gamma^2})q \\ & + \frac{\log 2}{\pi} I_h k^2 - r_h k(k+2) + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + o_h(q/\log q). \end{aligned} \quad (\text{FW29})$$

This use of the action is a receiving substitution, not an independent reproof of its arithmetic asymptotic. In the Gamma source the same proof of (FW27) gives the identical leading product; consequently

$$\mathcal{A}_2^m - \mathcal{A}_2^\Gamma = O_h(k + \log q), \quad \mathcal{A}_1^m - \mathcal{A}_1^\Gamma = \delta_{k,1}^{\text{ar}} + O_h(k + \log q). \quad (\text{FW30})$$

Here $\delta_{k,1}^{\text{ar}}$ is the original arithmetic baseline difference; the action's monic-window and penalty comparisons contribute only $O_h(k + \log q)$, as in CAI and AG28–29. Thus the $q/\log q$ arithmetic term is assigned to a specified source-correlation product.

6 The retained class's actual four-cutoff norm

The auxiliary zero argument is used only in W° , whose support avoids zero. For even $r = 2j$, the polynomial $(U_r^\circ(y)/U_r^\circ(0) - 1)/y$ has degree less than r and leaves residual $-U_r^\circ(y)/(yU_r^\circ(0))$ when approximating $-1/y$. This residual is orthogonal to every polynomial of degree less than r : for an odd monomial divide it by y and use orthogonality of U_r ; for an even monomial parity makes the integral zero. Therefore

$$H_{2j}^\circ(0) = H_{2j+1}^\circ(0) = M_{2j}^\circ/|U_{2j}^\circ(0)|^2. \quad (\text{FW31})$$

The equality at $2j + 1$ also follows because $C_{2j}^\circ(0) = 0$.

For completeness the return to the actual complex z has the finite estimate

$$\left| \log \frac{H_r^\circ(z)}{H_r^\circ(0)} \right| \leq 2(r+1)[- \log(1 - \varrho)] + 64(1 + R/d)\varrho, \quad 0 \leq r \leq q+1. \quad (\text{FW32})$$

For $r = 0$, use $H_0(z) = \text{Im}[-zM_0A_0]/d$. For even $r \geq 2$, write (FW6) as $|C_{r-1}|^2$ times the imaginary part of the first line of (FW14). The factor C_{r-1} is odd in its polynomial index, so (FW13) cancels its single factor z ; its ratio to the value at zero is A_{r-1} times the reciprocal of the individual squared-root product. Each root factor costs at most $-\log(1 - \varrho)$ in absolute log modulus. The imaginary-part ratio costs at most $16(R/d)\varrho$ on the guard. For odd r , the even C_{r-1} contributes $|z|^2$ and the second line of (FW14) contributes $\text{Im}(1/z) = d/|z|^2$; these cancel exactly. The same root and A estimates apply. The allowance in (FW32) bounds all these terms, including $r = 1$. This argument never treats a polynomial interpolation residual at nonzero z as an orthogonal projection.

The monic minima and (FW9) also give $(1 - \eta)\nu_j \leq \nu_j^\circ \leq \nu_j$. Combining (FW10)–(FW11), (FW26), (FW31)–(FW32) shows

$$\log \frac{E_{q-1}}{E_{2q-1}} = \log \nu_0 - \log \nu_q + 2 \log |U_q^\circ(0)| + O_h(1 + k^2/q). \quad (\text{FW33})$$

This endpoint computation needs no uniform moving-degree monic estimate. For each of the two specified monic minima, (FW7) bounds $\log(\nu_j^m/\nu_j^\Gamma)$ between $\log \ell$ and $\log u$; their difference is therefore within Λ . In Gamma, AG19 with $(n, s) = (q/2, 0), (q/2 + 1, 1)$ and AG22 give directly

$$\begin{aligned} \log \nu_q^\Gamma &= 4q \log q + (F - L_{\text{eq}} + 3)q + O_h(\log q), \\ \log \nu_0^\Gamma &= 2q \log q + (4 \log 2 - 2 \log \pi - 2)q + O_h(\log q). \end{aligned} \quad (\text{FW34})$$

Indeed the even relation block acquires just its next monic vector; its determinant ratio is $H_{q/2+1}^q/H_{q/2}^q$. Subtracting AG19 at these two literal block sizes produces the first line, with its qL_{eq} term and all powers of q retained. Thus (FW33) equals $[2L_{\text{eq}} - F - 5 + 4 \log 2 - 2 \log \pi]q + O_h(k + \log q)$. The original ECL/LRS16 convention states $C_\partial = 4 \log(4/\pi) - 10 - 2F + 4L_{\text{eq}}$, so this is $C_\partial q/2 + O_h(k + \log q)$. Finally $r = 0, q$ are even, and (FW11), (FW16) bound each corresponding T_r by $O_h(k^2/q^2)$. Returning through (FW4) proves

Theorem 3 (Both original endpoint windows).

$$\log \frac{E_{q-1}}{E_{2q-1}} = \frac{C_{\partial q}}{2} + O_h(k + \log q), \quad \log \frac{E_q}{E_{2q}} = \frac{C_{\partial q}}{2} + O_h(k + \log q). \quad (\text{FW35})$$

In particular

$$\boxed{\log \frac{E_{q-1}E_q}{E_{2q-1}E_{2q}} = C_{\partial q} + O_h(k + \log q).} \quad (\text{FW36})$$

All occurrences of $|b_\lambda|^2$ cancel between the same original class at different cutoffs; they were retained in (FW4) before that cancellation. This proof evaluates the actual original-source class.

7 Exact receiving map into the original observation quotient

Retain the actual fixed observation $\Lambda : E \rightarrow B = \text{im } \Lambda$, including its period map, and an injective frame I for $K = \ker \Lambda$. Define

$$P_N = I(I^*G_N I)^{-1}I^*G_N, \quad Q_N = (\Lambda G_N^{-1}\Lambda^*)^{-1}, \quad \theta_N = v^*G_N P_N v / E_N. \quad (\text{FW37})$$

The exact minimum lift $G_N^{-1}\Lambda^*Q_N : B \rightarrow E$ is a right inverse of Λ , is orthogonal to K , and proves

$$G_N(I - P_N) = \Lambda^*Q_N\Lambda, \quad e_{B,N} := (\Lambda v)^*Q_N\Lambda v = E_N(1 - \theta_N). \quad (\text{FW38})$$

This supplies the typed metric map between the just evaluated class norm and its observed norm. Its complete proof is the same quadratic minimum as ORE2–3 [9]; the general minimum framework is credited there to Boyd and Vandenberghe [4, Appendix C.4.1].

If $\Lambda v \neq 0$, strict positivity of Q_N gives $1 - \theta_N > 0$ at every finite cutoff. Equations (FW35)–(FW38) then yield the new receiving formula

$$\boxed{\log \frac{e_{B,q-1}e_{B,q}}{e_{B,2q-1}e_{B,2q}} = C_{\partial q} + \log \frac{(1 - \theta_{q-1})(1 - \theta_q)}{(1 - \theta_{2q-1})(1 - \theta_{2q})} + O_h(k + \log q).} \quad (\text{FW39})$$

The unresolved part in this specific receiver is now four actual endpoint projection angles. It has not been replaced by ranks or by a scalar total. If $\Lambda v = 0$, the observed energy is identically zero and (FW38), rather than an undefined logarithm, is the receiving statement.

There is also a positive exact calculation at each original increment. With $b = b_{N+1}$ and $\beta = b^*G_N b$, put $\alpha_N = b^*G_N P_N b / \beta$, $d_{K,N} = \det(I^*G_{N+1}I) / \det(I^*G_N I)$, and $V_N = b^*G_N P_N v / (b^*G_N v)$. The original source increment and its restriction, checked by multiplying the proposed inverse, give

$$G_{N+1}^{-1} = G_N^{-1} + b b^* / \omega_{N+1}, \quad d_{K,N} = 1 - (1 - \Delta_r)\alpha_N, \quad \frac{e_{B,N+1}}{e_{B,N}} = 1 - \frac{T_r |1 - V_N|^2}{d_{K,N}(1 - \theta_N)}. \quad (\text{FW40})$$

The quotient update denominator is exactly $\omega_{N+1} + \beta(1 - \alpha_N) = (\omega_{N+1} + \beta)d_{K,N}$; pairing the inverse update with Λv proves (FW40). The matrix inverse method is Hager’s rank-one update [5], used with the original vector and denominator. Combining with $E_{N+1}/E_N = 1 - T_r$ gives the undivided identity

$$(1 - T_r)(1 - \theta_{N+1}) = 1 - \theta_N - T_r |1 - V_N|^2 / d_{K,N}.$$

Thus the evaluated contraction (FW36) and all actual observation responses belong to this same sequence of attained minima. For any positive sequence a_N , cancellation proves $\log(a_{q-1}a_q / (a_{2q-1}a_{2q})) = -\sum w_r \log(a_{N+1}/a_N)$. Applying it to (FW40) recovers (FW39) with the exact endpoint weights. It retains the finite common scalar phase correction in the three signed currents; their sum still uses $P_K + P_B + P_\times = 1$ in the original response-product notation. No individual signed current is assigned here.

8 Degree-resolved contraction of the original class

Fix $s \in (0, 1]$, put $r = \lfloor sq \rfloor$, $n = \lfloor r/2 \rfloor$, and retain the equilibrium for $V_s(x) = (\pi/s)\sqrt{x} - (2/s)\log x$. Write $L_s = \mathcal{L}(2/s, \pi/s)$, $F_s = -\min \mathcal{E}_{2/s, \pi/s}$ and $\lambda_s = \lambda(2/s, \pi/s)$, with the original Robin convention $V_s - 2 \int \log |x - y| d\rho_s(y) = \lambda_s$ on its support. The comparison proof (FW20)–(FW23) applies to this n without change: its common spaces still have codimension one, all output degrees remain at most $2q + 2$, and its rational comparison retains the same B_m, B_Γ . Choose the fixed gap below the support of this equilibrium as well. For fixed s , this is a change of an auxiliary estimate only; (FW10) returns its explicitly bounded contribution to the same original minimum.

For the Gamma calculation let $b_s = sq - 2n \in [0, 2)$. Its exact additional analytic factor, on top of $-3 \log x/4 - \tau_k/x$, is

$$\frac{b_s}{s} \log x - \frac{\pi b_s}{2s} \sqrt{x}.$$

It is retained in both Gram determinants. The analytic-statistic estimate used in (FW25) is uniform for this bounded pair (b_s, τ_k) : the logarithmic determinant minus its linear statistic is convex in the pair; bound it at the vertices and centre of a fixed larger rectangle and use the same convex-combination argument as for (FW25). Thus

$$\log |U_{2n}^\circ(0)| = 2n \log q + nL_s + O_{h,s}(k + \log q). \quad (\text{FW41})$$

Both complementary-region bounds used in AG have the same proof here, with $D/q \leq 1 + o(1)$ and a fixed positive s . Their constants may depend on s ; no uniformity as $s \downarrow 0$ is used in (FW41).

The original Gamma monic norm of degree $2n$ is the ratio of its even Gram blocks of sizes $n + 1, n$. To use a fixed potential in both, the larger block's analytic factor is the smaller one's factor plus V_s . The leading log determinant difference is therefore $2nF_s + n \int V_s d\rho_s = -n\lambda_s$; the equality follows by integrating the Robin equation. The exact row, column and mass powers of q contribute $(2q + 4n + 1/2) \log q$. Comparison of the actual monic minima by (FW7) consequently gives

$$\log \nu_{2n} = (2q + 4n) \log q - n\lambda_s + O_{h,s}(k + \log q). \quad (\text{FW42})$$

The term $(1/2) \log q$ has only been absorbed into the stated logarithmic error, not multiplied by a Gram rank. Combine (FW41)–(FW42) with (FW31)–(FW32), and use $\log \nu_0 = 2q \log q + (2 \log(4/\pi) - 2)q + O_h(k + \log q)$ from (FW34). If r is odd, its extra even-index update changes the logarithm by $O_h(k^2/q^2)$, by (FW16). We obtain, at every fixed multiplicity,

$$\boxed{\begin{aligned} -\log \frac{E_{q-1+\lfloor sq \rfloor}}{E_{q-1}} &= q\mathcal{J}(s) + O_{h,s}(k + \log q), \\ \mathcal{J}(s) &= s(L_s + \lambda_s/2) + 2a_0 - 2, \quad a_0 = \log(4/\pi). \end{aligned}} \quad (\text{FW43})$$

This is the original degree profile, with an exact expression for its derivative. In the conventions of LRS10–15 [10], let

$$\psi(t) = (1+t)(1 - \log(1+t)) - \frac{t}{4}\lambda_t, \quad f(t) = 2\psi(t) - 2(t+1)\psi'(t).$$

For the original elliptic modulus κ_t , write $r_t = \sqrt{1 - \kappa_t^2}$. The endpoint equation and its consequences are

$$\frac{E(\kappa_t)}{r_t K(\kappa_t)} = 1 + t, \quad \psi(t) = \log \frac{1 + r_t}{E(\kappa_t)} + t \log \frac{\kappa_t}{E(\kappa_t)}, \quad f(t) = \log \frac{1 + r_t}{1 - r_t} > 0. \quad (\text{FW44})$$

Here K, E are complete elliptic integrals; the modulus is unrelated to the observation kernel. The derivative identity $\psi' = \log(\kappa_t/E(\kappa_t))$ follows from the original endpoint variation, as proved

in LRS8–12. Its convergent small-modulus expansion gives $\kappa_t = 2t^{1/4}(1 + O(\sqrt{t}))$, hence $f(t) = -\frac{1}{2} \log t + O(\sqrt{t})$, an integrable function at zero. No asymptotic remainder has been differentiated.

For an explicit check of the integrated identity, LRS15 gives the continuous primitive $\mathfrak{F}(t) = t^2 F_t/4 + (1+t)^2(3/2 - \log(1+t))/2 - 3/4$ of ψ , with value zero at zero. Integration by parts gives

$$\int_0^s f(t) dt = 4\mathfrak{F}(s) - 2(s+1)\psi(s) + 2a_0.$$

Substitution and the proved equilibrium equality $F_s = -\lambda_s/2 - (2/s+1) + L_s/s$ reduce this expression exactly to $s(L_s + \lambda_s/2) + 2a_0 - 2$. Thus $\mathcal{J}(s) = \int_0^s f$, including its continuous zero endpoint.

Put $g_r = H_{r+1}/H_r = 1 - T_r$. The finite positive measures

$$\mu_k = \frac{1}{q} \sum_{\substack{0 \leq r < q \\ r \text{ modd}}} (-\log g_r) \delta_{r/q} \quad \text{converge weakly on } [0, 1] \text{ to } f(s) ds. \quad (\text{FW45})$$

Indeed (FW16) bounds the sum of all even-index losses by $O_h(k^2/q) = o(q)$. The full prefix sum is exactly $-\log(H_{\lfloor sq \rfloor}/H_0)$, whose quotient by q tends to $\mathcal{J}(s)$ by (FW43). For any fixed $s > 0$, sandwiching its boundary atom between prefixes at $s - \epsilon$ and $s + \epsilon$ proves convergence at s by continuity of $\mathcal{J}$. At zero, total prefix mass is at most the mass below any positive ϵ , whose limit tends to zero. At one, (FW35) gives the full mass. These statements prove weak convergence against continuous functions by step-function approximation. They do not claim a pointwise limit of any individual odd removal fraction.

9 Original conductor bound at every cutoff and its determinant return

This section retains the original simple-quartet conductor domain: $m_0 = 1$, $k \geq 9$, its fixed actual period ϖ , and the finite root guards in FEX/LRS. Its full original observation has $K = \ker \Lambda_k \subset \{[P] : \chi_{k-8} \mid \mathcal{T}_A P\}$, with

$$\begin{aligned} q' &= (k-7)^2, \quad \Delta_A = q - q' = 16k - 48, \quad c' = c - 4, \\ \mathcal{T}_A P(S') &= \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad \beta_{8,a,b} = 4 + (2a-8)\delta + i(2b-8)\gamma, \\ \mu_j &= \sum a_{ab} \beta_{8,a,b}^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80, \quad \mathcal{C}_A = (v!/\mu_v) \mathcal{T}_A. \end{aligned} \quad (\text{FW46})$$

All its constants and its scalar are the original ones, as reproduced in the full conductor proof [11]. At a lower root, every shifted point is an upper root. Consequently $\mathcal{T}_A(\chi H)$ vanishes at every lower root and is divisible by the complete simple lower polynomial. Thus the conductor map is defined on the original quotient and factors through the actual observation; this proves its use on every minimum representative, rather than only on a chosen coefficient section. The leading coefficient of $\mathcal{C}_A P$ in degree $N - v$ is $(N)_v$ times the leading coefficient of P , by the finite Taylor formula.

The forward Gamma bound retains the centre displacement exactly:

$$\|\mathcal{C}_A P\|_{\sigma, c-4} \leq M_N \|P\|_{\sigma, c}, \quad M_N = \max \left\{ 1, \frac{v!}{|\mu_v|} \sum |a_{ab}| e^{|\beta_{8,a,b}-4|(\mathbf{H}_N+2)} \right\}, \quad \mathbf{H}_N = \sum_{j=1}^N \frac{1}{j}.$$

The full derivation in [11] expands differentiation in the Meixner–Pollaczek basis and bounds its weighted row and column sums by $\mathbf{H}_N$ and $\mathbf{H}_N + 4$, respectively. Exponentiating the finite nilpotent derivative gives exactly this translation bound at every finite degree N , so $\log M_N = O_{h,\varpi}(\log q)$ on the window.

Let $\nu_{j,1}^-$ be the full monic lower-relation minimum in the order-one Gamma source. For an arbitrary $x \in K$, write its actual minimum representative as $T_N x = P$ and let its leading coefficient be $a_N(P)$. The divisibility just proved and (FW7) imply

$$\|x\|_{G_N}^2 \geq \ell M_N^{-2}(N)_v^2 \nu_{N-v-q',1}^- |a_N(P)|^2.$$

This is also true if the leading coefficient is zero. The exact boundary pairing is $b_N^* G_N x = \omega_N a_N(P)$. Take its squared dual norm on the entire K , use $\omega_N \leq u \gamma_N$, and retain $F_{11} = \omega_N \eta_N$, with $\eta_{q-1} = 1$ and $\eta_N = 1 - \Delta_{N-q}$ for $N \geq q$. This proves, throughout the window,

$$\pi_{1,K,N} := \frac{\|P_N b_N\|_{G_N}^2}{F_{11}} \leq \min \left\{ 1, \frac{u}{\ell} \frac{M_N^2 \gamma_N}{\eta_N(N)_v^2 \nu_{N-v-q',1}^-} \right\}, \quad q-1 \leq N \leq 2q. \quad (\text{FW47})$$

Its lower-relation degree is nonnegative on the original eventual guards. The inherited LRS1 proves the upper degree bound $\leq 2q'$ there. All finite lower endpoints LRS4 may be inserted for the displayed monic norm before taking asymptotics; hence (FW47) has an explicit finite receiver containing no observation determinant.

For the rate use LRS3–11, in the order-one reference for both complete simple root lists. It gives, uniformly for $0 \leq j \leq 2Q$,

$$\begin{aligned} \log(\nu_{j,1}^{(b)}/\gamma_{Q+j}) &= 2Q\psi(j/Q) + O_h(\log q), \\ Q &= (b+1)^2, \quad b = k, k-8. \end{aligned}$$

The original arithmetic monic norms and relation norms differ from their Gamma values by log factors in $[\log \ell, \log u]$; thus their ratio costs at most $\Lambda = O_h(k + \log q)$. The exact boundary-radius formula preceding (FW19), (FW16), and the current identity used in (FW28) now give

$$-\log p_{1,q-1+r} = 2q\psi(r/q) + O_h(k + \log q), \quad 0 \leq r \leq q+1. \quad (\text{FW48})$$

For clarity, $\log \kappa_N^2$ equals $\log(\nu_r/\omega_{q+r}) + \log(\omega_{N+1}/\omega_N)$ plus the two actual $\log(1 - \Delta)$ factors. Comparison with $\gamma_{n+1}/\gamma_n = (n+1)(n+1/2)$ bounds that recurrence logarithm by $O_h(k + \log q)$; (FW19) bounds the remaining factors. Also (FW11), (FW16) and $p_2 \leq 1$ give $|\log p_2| = O_h(\log q)$. These observations prove (FW48) including $r = 0$.

Set $F_*(n, r) = 2n\psi(r/n)$ to avoid confusion with the equilibrium constant F . Its exact derivative difference is $\partial_n F_* - \partial_r F_* = f(r/n)$, by (FW44). Therefore

$$F_*(q, r) - F_*(q', r + \Delta_A) = \int_0^{\Delta_A} f\left(\frac{r+s}{q-s}\right) ds.$$

The integral at $r = 0$ is convergent. A shift of the second argument by the fixed $v+1$ costs $O_h(\log q)$ using the proven logarithmic endpoint bound for ψ' . Since $f(t) \leq -\frac{1}{2} \log t + C$ for $t \leq 1$ and is bounded on the remaining compact interval, its integral is at most $\frac{\Delta_A}{2} \log^+(q/(r + \Delta_A)) + C\Delta_A$. The bound follows by integrating $\log(q/(r+s))$ exactly; its excess over $\Delta_A \log(q/(r + \Delta_A))$ is $\Delta_A - r \log(1 + \Delta_A/r) \leq \Delta_A$. Finally the monic factorial factor satisfies the exact identity

$$\frac{\gamma_N}{\gamma_{N-v}(N)_v^2} = \prod_{j=0}^{v-1} \frac{N-j-1/2}{N-j} \leq 1.$$

Using it in (FW47) and then dividing by (FW48) gives

$$|U_N| \leq \sqrt{\theta_N} \exp \left\{ \frac{\Delta_A}{4} \log^+ \frac{q}{r + \Delta_A} + C_{h,\varpi} k \right\}, \quad r = N+1-q. \quad (\text{FW49})$$

Indeed $|U_N|^2 \leq \theta_N \pi_{1,K,N}/p_{1,N}$ is precisely Cauchy–Schwarz on the two actual K projections. Similarly $|V_N|^2 \leq \theta_N/p_{2,N}$ and (FW16) yield

$$|V_N| \leq C_h \sqrt{\theta_N} \begin{cases} q/k, & r \text{ even,} \\ 1, & r \text{ odd.} \end{cases} \quad (\text{FW50})$$

For odd r , its lower bound follows explicitly from $b_r \geq Y^2/\widehat{C}^2$; no gap below one is asserted.

There is a further direct determinant receiver. The original rank-one update (FW40), now applied to $b = b_{N+1}$ in K , gives $\pi_{1,K,N+1} = \Delta_r \alpha_N / d_{K,N}$. Solving this equality with the displayed value of $d_{K,N}$ yields

$$d_{K,N} = \frac{\Delta_r}{\Delta_r + (1 - \Delta_r)\pi_{1,K,N+1}}, \quad 0 \leq -\log d_{K,N} \leq \frac{\Delta_A}{2} \log^+ \frac{q}{r + \Delta_A} + C_{h,\varpi} k. \quad (\text{FW51})$$

For the inequality repeat (FW47) at $N + 1$ and compare its monic norm with $-\log \Delta_r = 2q\psi(r/q) + O_h(k + \log q)$. The lower index is now $r + \Delta_A - v$, so the identical integral estimate applies with the fixed shift v . The logarithm $\log(1 + e^x)$ costs at most $\log 2 + \max(x, 0)$; all constants are absorbed in the specified $C_{h,\varpi} k$. This is an inequality for the same original restriction determinant, rather than an allocation from its rank.

In the complete weighted window, comparison with $\int_0^q \log(q/x) dx = q$ gives

$$0 \leq \log \frac{\det(I^* G_{q-1} I) \det(I^* G_q I)}{\det(I^* G_{2q-1} I) \det(I^* G_{2q} I)} = \sum_{r=0}^q w_r [-\log d_{K,q-1+r}] \leq C'_{h,\varpi} k q. \quad (\text{FW52})$$

This carries the established kq control into the literal observation kernel through its actual conductor. It does not equate this object with another native flag. The exact morphism to the observation quotient is $\det Q_{N+1} / \det Q_N = \Delta_r / d_{K,N}$ from (FW40); its full-window loss is the already evaluated original total minus the exact restriction loss in (FW52). The latter bound does not specify its order- kq coefficient.

10 Precise integration scope

Equations (FW27)–(FW30) replace the low-cutoff-only source-product status in the preceding edition. Equations (FW35)–(FW36) evaluate its previously unevaluated full terminal-class norm return. Equations (FW38)–(FW40) push those results into ORE’s original observation, including the full conductor cross blocks retained in COI11–13 and ORE19. The exact finite term D_k in (FW18) improves the incoming proof’s dependency: an exponential estimate for Δ_r is unnecessary for those conclusions. Equations (FW41)–(FW45) prove the further fixed-degree-fraction profile and weak loss measure. Equations (FW46)–(FW52) prove the complete-window conductor response bound on its original simple-quartet domain and carry it into the original restriction determinant. The uniform moving-degree first-energy estimate (FW48) is certified here on that same simple domain; the main product and class-profile theorems retain every fixed multiplicity. The seven native determinant allocations and projected signs retain their actual period-dependent projections. No claim to an RH proof is made.

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